Homework 4

Due date: Apr.20^{th}

Turn in your homework in class

Rules:

- Please work on your own and submit **on time**. Discussion is permissible, but extremely similar submissions will be judged as plagiarism!
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain **3** decimal places if rounding is required without special requirements for the question,
- Please staple your homework before submitting; take a photo and upload it on time for online record, including the grade (with any possible deductions) after receiving the marked copy.

1

[12 points] The circuit is shown in **Figure 1**. The switch has been closed long enough before $t = 0$s, and it is opened at $t = 0$s. $R_1 = 8\Omega$, $R_2 = 10\Omega$, $C_1 = 4$F, $L_1 = 3$H. Determine the voltage $v(t)$ of C_1.

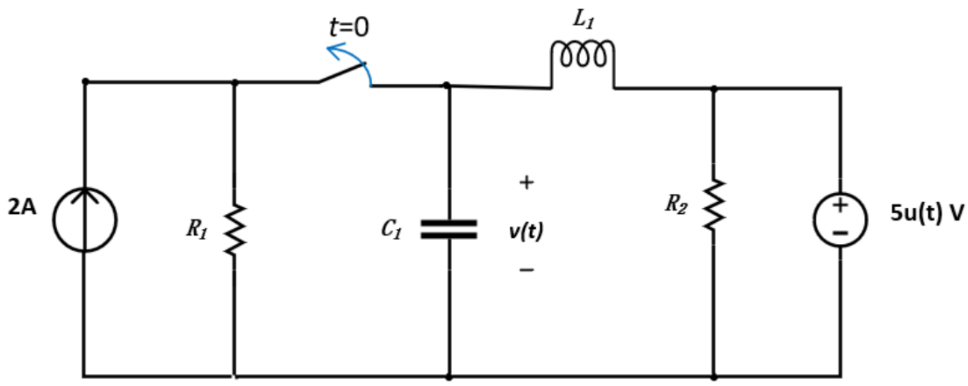

Figure 1

2

[14 points] In the Op Amp circuit of **Figure 2**, $V_{\text{in}} = 3u(t)$ V, $R_1 = 15$ kΩ, $R_2 = 20$ kΩ, $C_1 = 100$ μF, $C_2 = 200$ μF. The Op Amp is ideal and operates in its linear region. Determine $V_{out}(t)$ for $t > 0$.

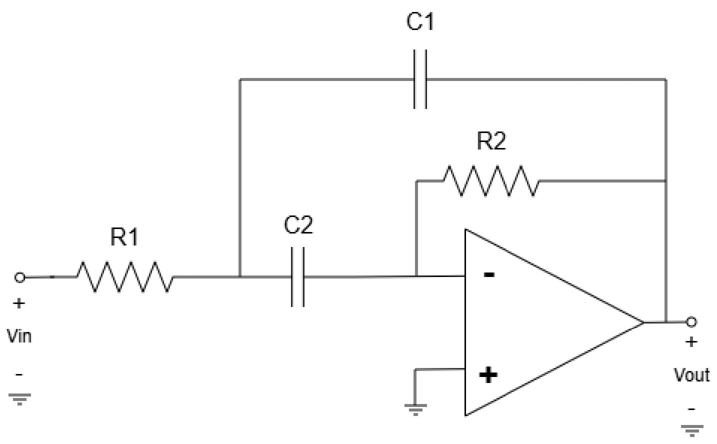

Figure 2

3

[12 points] The circuit is shown in **Figure 3**. The circuit has been in state 1 for a long time. At $t = 0$, it switches to state 2.

(a) Please calculate: v(0^+), $\frac{dv(0^+)}{dt}$;

(b) When $t \geq 0$,Find v(t).

(c)If we wish to change the resistance value (0.5 Ohm) to turn this circuit into a critically damped state, determine the required resistance. In this case , find v(t) $t \geq 0$.

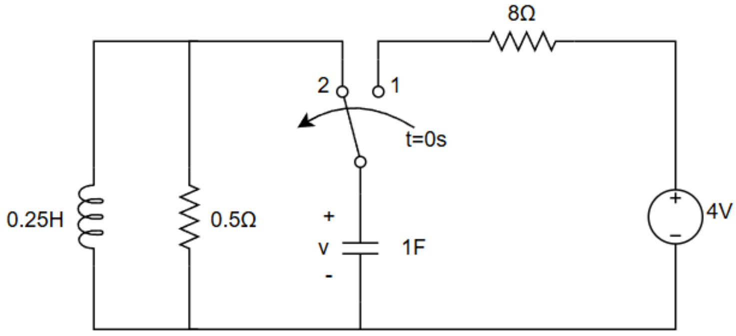

Figure 3

4

[16 points] The circuit is shown in **Figure 4**. Known that $I_1(t) = 2u(t)$ A, $I_2(t) = 2u(t)$ A, $R_1 = 20\Omega$, $R_2 = 10\Omega$, $R_o = 10\Omega$, $L_1 = 100$ mH, $C_1 = 1$ mF. S_1 and S_2 are both turned off and the circuit has already reached the steady state for $t < 0$ s. Suppose two switches can turn on or turn off immediately.

(a). At $t = 0$ s, S_1 turns on and S_2 turns off. Find the voltage V_o for $0\,\text{s} \le t < 10^{-2}\,\text{s}$.

(b). At $t = 10^{-2}$ s, S_1 turns off and S_2 turns on. Find the voltage V_o for $t \le 10^{-2}\,\text{s}$.

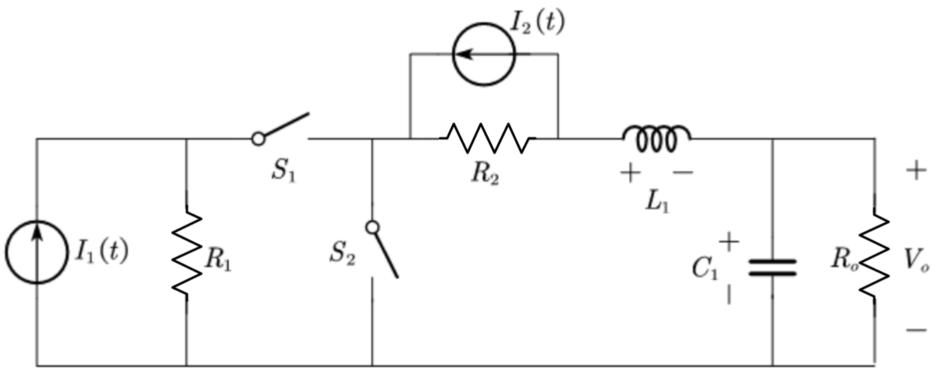

Figure 4

5

[16 points] The circuit is shown in **Figure 5**. Known that $L_1 = 0.8\text{H}$, $C_1 = 0.5\text{F}$, $V_1(t) = 12\text{V}$ ($t \geq 0\text{s}$), $I_1(t) = 20\text{A}$ ($t \geq 0\text{s}$), $I_2(t) = 5\text{A}$ ($t \geq 0\text{s}$), $I_3(t) = 2\text{A}$ ($t \geq 0\text{s}$), $R_1 = 2\Omega$, $R_2 = 4\Omega$, $R_3 = 4\Omega$, $R_4 = 6\Omega$. The energy stored in the capacitor at $t = 0\text{s}$ is 0.08J and there is no energy stored in the inductor.

Determine the response of $V_{C1}(t)$ for $t \geq 0\text{s}$. (Hint: You can first find the Thevenin equivalent circuit at terminals A and B (excluding the capacitor C_1 and the inductance L_1). In this case, the overall circuit can be simplified as a second order RLC circuit.)

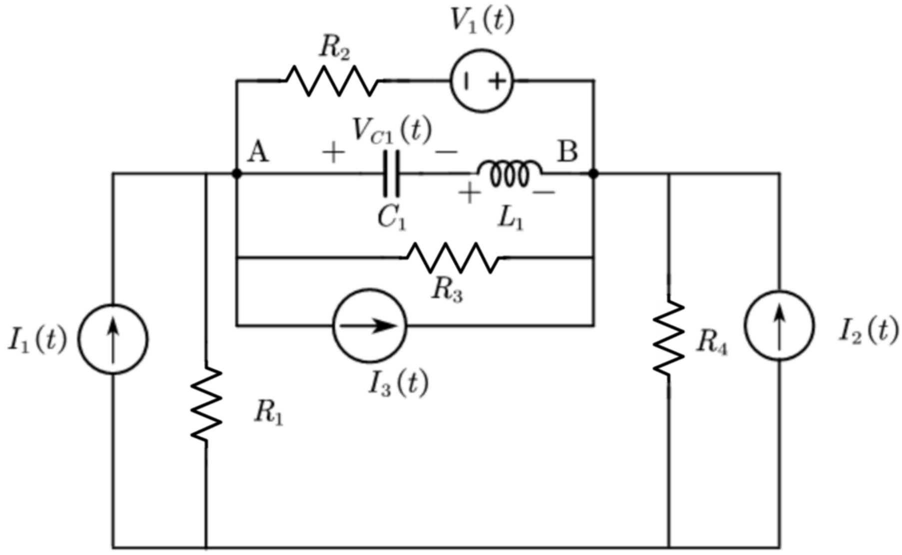

Figure 5

6

[14 points] For the circuit in **Figure 6**, the Op Amp is ideal and operates in its linear region. Find out $v_C(t)$ as $v_C(0-) = -35\text{V}$ and $i_L(0-) = 1\text{A}$.

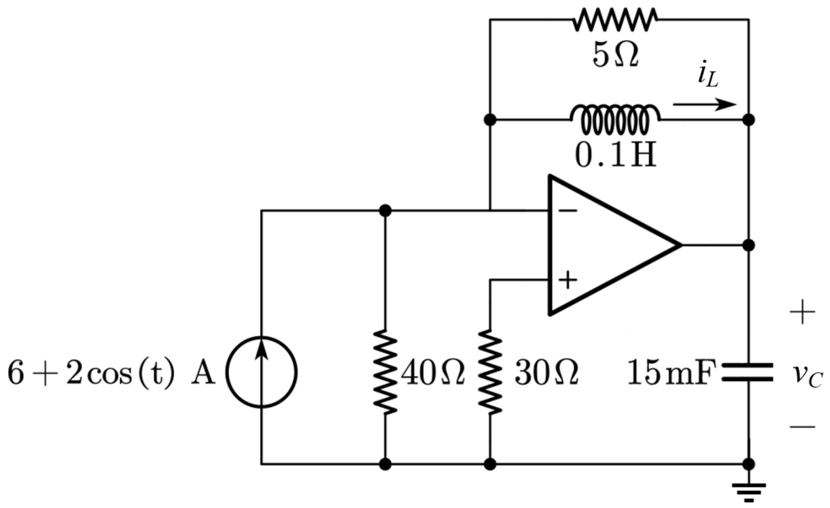

Figure 6

7

[16 points] For the circuit in **Figure 7**, the expression $i_s(t)$ for the current source follows:

$$i_s(t) = \begin{cases} (t+1)e^{-3t}, & 0 < t \leq 1, \\ e^{-2t+1}\big(2\cos(3t) + \sin(3t)\big), & t > 1. \end{cases}$$

Assume the system reaches steady state before $t = 0$.Find $i(t)$ and $v(t)$ for $t > 0$.

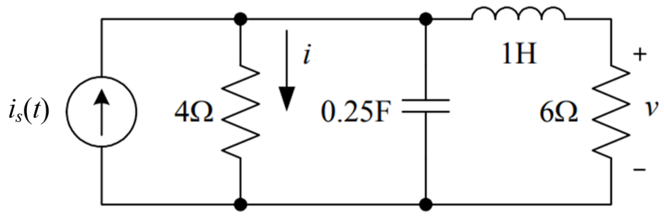

Figure 7